

Immunology (PPTz)

- (Merged by Talha Waraich
- For help What's App or Telegram at
- 03488485459)

Immunology

Overview of the
Immune System as
the Body's main
Defense
Mechanism

Immunology

Immunology,
Immunity , Immune
System & Immune
Response

Terminologies

- Immunity
- Immune System
- Immune Response
- Immunology

Immunology

Infectious vs
Non-Infectious
diseases

Infections Vs Non-Infections

Infections

- Contagious
- Transmissible
- Communicable
- Pathogens (Bacteria, Virus, Fungus & Parasites) *e.g* Hepatitis

Non-Infections

- Non-Communicable
- Metabolic disorders
- Genetic Factors *e.g* Diabetes Mellitus

Immunology

Importance of Immune System

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03488485459)

Importance of Immune System

Role of Immune System

- Defense against infections: AIDS
- Responsive against tissue grafts & newly introduced proteins: Barrier for tissue transplantation & gene therapy

Importance of Immune System

Role of Immune System

- Surveillance against cancers (Tumors)
- Immune Products *e.g* Antibodies use in diagnostics & therapeutics

Immunology

Difference between Innate & Adaptive Immune system

Difference B/W Innate & Adaptive Immune System

Two (02) types of Immunity

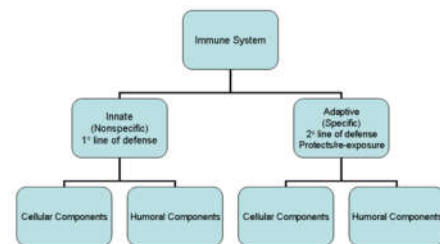
Innate Immune system

- Non-Specific
- First line of defense
- Readily available

Adaptive Immune System

- Specific
- Second line of Defense
- Needs to be activated

Difference B/W Innate & Adaptive Immune System



Difference B/W Innate & Adaptive Immune System

Differences B/W Innate & Adaptive Immune System

Innate Immune System	Adaptive Immune System
Response is antigen-independent	Response is antigen-dependent
There is immediate maximal response	There is a lag time between exposure and maximal response
Not antigen-specific	Antigen-specific
Exposure results in no immunologic memory	Exposure results in immunologic memory

Immunology

Components of Innate Immune System

Components of Innate Immune System

Three (03) components or barriers of Immunity against infections

- Anatomical (Physical) barriers
- Humoral (Secretory) barriers
- Cellular barriers

Immunology

Physical barriers of Innate Immune system

Physical Barriers of Innate Immune System

Anatomical Barriers

Mechanical Factors

- Skin
- Epithelium membrane (Squamous)
- Desquamation (Flushing of microbes)
- Peristalsis/ Ciliary movement (GIT)
- Flushing actions of tears & saliva
- Trapping actions of mucus lining of respiratory & GIT

Physical Barriers of Innate Immune System

Anatomical Barriers

Chemical Factors

- Fatty acids in skin
- Lysozymes & Phospholipases in tears, saliva, nasal secretions
- Defensins show antimicrobial peptides (GIT/Lungs)
- Low pH of sweat & GIT secretions
- Surfactants act as opsonins (Lungs)

Physical Barriers of Innate Immune System

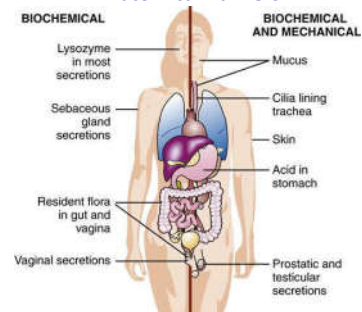
Anatomical Barriers

Biological Factors

- Normal flora of skin & GIT
- Prevents the colonization of bacteria
- By secreting toxins against foreign microbes
- Physical potential for utilization of nutrients by competing the pathogenic bacteria

Physical Barriers of Innate Immune System

Anatomical Barriers



Immunology

Secretory molecules of Innate Immune System

Secretory Molecules of Innate Immune System

Humoral barriers

- After breaching anatomical barriers, infectious agent penetrate the deep tissues
- Inflammation: tissue response
- Humoral (Secretory) barriers mediate the inflammatory process
- Signs of inflammation: pain, heat, redness, swelling & loss of tissue functions

Secretory Molecules of Innate Immune System

Humoral barriers

1) Complement System

- Complement Proteins
- Plasma proteins
- After activation, mediate the lysis of infectious agent

2) Coagulation System

- Coagulation Factors
- Plasma Proteins
- Chemotactic agent
- Antimicrobial activity e.g β -Lysins

Secretory Molecules of Innate Immune

Humoral barriers

3) Lactoferrin & transferrin

- Plasma proteins
- Ability to bind iron
- Deprive bacteria from iron

4) Interferons

- Plasma proteins
- Inhibit viral replication in cells

Humoral barriers

5) Lysozymes

- Present in saliva, nasal secretions & tears
- Has ability to lyse bacterial cell wall after digesting peptidoglycan

6) Interlukin-1 (IL-1)

- Cytokine
- Released by immune cells after activation
- Induce fever & antimicrobial acute phase proteins

Cells of Innate Immune system

Cells of Innate Immune System

Cellular barriers

- In inflammatory process during infections, under the action of various humoral substances immune cells recruit towards the site of infection (Chemotaxis)
- Immune cells from blood
- Immune cells from the inflamed tissues

Cells of Innate Immune System

Cellular barriers

1) Neutrophils

- Polymorphonuclear leucocytes (PMNs)
- Phagocytose the invading agent & kill intracellularly
- Immune cells from the inflamed tissue



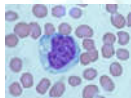
<http://www.microbiologybook.org/ghaffar/neutrophil.jpg>

Cells of Innate Immune System

Cellular barriers

2) Macrophages

- Phagocytic cells
- Tissue macrophages: Histocytes
- Circulating macrophages: Monocytes
- Involved in phagocytosis & intracellular killing



<http://undergraduate.vetmed.wsu.edu/images>

Cells of Innate Immune System

Cellular barriers

3) Natural Killer (NK) Cells

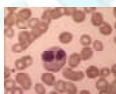
- Have the ability to kill viral infected cells non-specifically
- Also kill transformed or Tumorous cells
- Role in tumor surveillance

Cells of Innate Immune System

Cellular barriers

4) Eosinophils

- Blood Granulocytes
- Contain granules which are effective against parasitic infections
- Also cause the cytotoxicity of parasitic infected cells via receptors non-specifically



<http://www.microbiologybook.org/ghaffar/eosinophil.jpg>

Phagocytosis

(Merged by

Talha Waraich

For help What's App or Telegram at

03488485459)

Phagocytosis

Phagocytosis

- The process of engulfing the invading or infectious agent by phagocytes
- Professional phagocytes
- Polymorphonuclear Phagocytes e.g Neutrophils
- Mononuclear Phagocytes e.g monocytes, histocytes, Kupffer cells

Phagocytosis

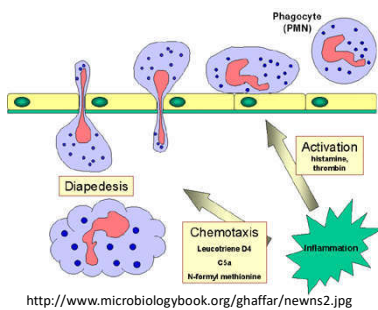
Process of Phagocytosis

1) Chemotaxis

- Chemo: Chemical, Taxis: Movement
- SOS signals from bacteria
- Secretory molecules like coagulation peptides, complement
- Migration of phagocytes across the capillary wall (Diapedesis)

Phagocytosis

Chemotaxis of Phagocytes



Phagocytosis

Process of Phagocytosis

2) Attachment

- Phagocytes carry receptors for binding with bacterial surface components
- Fc receptors
- Complement receptors
- Scavenger receptors
- Toll like receptors (Pattern Recognition Receptors)

Phagocytosis

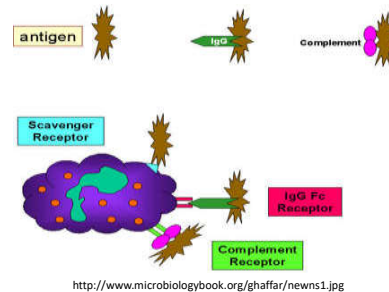
Process of Phagocytosis

2) Attachment

- On bacterial surface phagocyte's receptors interact with following components
- Opsonization
- IgG (Opsonins)
- Complement proteins
- Pathogen Associated Molecular Patterns (PAMPs) e.g LPS, flagellin etc

Phagocytosis

Attachment of Bacteria to Phagocytes



Phagocytosis

Process of Phagocytosis

3) Phagosome Formation

- Once bacteria attach with its corresponding receptor on phagocyte
- Pseudopod extends around the bacterium to form a vesicle called as Phagosome
- Phagosome contains the trapped bacteria inside

Phagocytosis

Process of Phagocytosis

4) Phagolysosome Formation

- Phagosome with trapped bacteria fuse with secretory vesicles of phagocytes called "Lysosomes"
- Phagolysosome formation result in the action of hydrolytic enzymes i.e Lysozyme

Immunology

Intracellular killing

Intracellular killing

Intracellular Killing

- After phagocytosis the ingested bacteria is being killed by a process called as Intracellular Killing
- Two ways of intracellular killing
- Oxygen independent
- Oxygen dependent

Intracellular killing

Intracellular Killing

1) Oxygen Independent

- No need of oxygen for such kind of intracellular killing of bacteria
- Granules & Vesicles of phagocytes secrete hydrolytic proteins
- Those proteins are bacteriocidal in nature according to their modes of action

Intracellular killing

Mechanisms of Oxygen Independent Killing

Effector Molecule	Function
Cationic proteins (including cathepsin)	Damage to microbial membranes
Lysozyme	Splits mucopeptide in bacterial cell wall
Lactoferrin	Deprives proliferating bacteria of iron
Proteolytic and hydrolytic enzymes	Digestion of killed organisms

Intracellular killing

Intracellular Killing

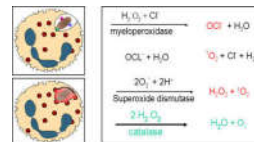
2) Oxygen Dependent

- Requirement of oxygen for such kind of intracellular killing of bacteria
- Also called "Respiratory Burst" as requirement of glucose & oxygen increased after phagocytosis
- Oxygen containing bacteriocidal radicals are produced

Intracellular killing

Intracellular Killing

2) Oxygen Dependent



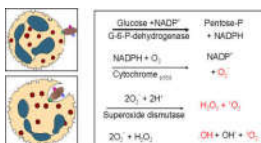
<http://www.microbiologybook.org/ghaffar/ns2000-3a.jpg>

- Myeloperoxidase (MPO) dependent
- MPO from granules of phagocytes
- Halide ions (OCl^-) are formed which are bacteriocidal

Intracellular killing

Intracellular Killing

2) Oxygen Dependent



<http://www.microbiologybook.org/ghaffar/ns2000-3.jpg>

- Myeloperoxidase independent
- Involvement of Hexose monophosphate shunt
- Reactive Oxygen Species (ROS) e.g Superoxide radicals, hydrogen peroxide & singlet oxygen

Immunology

Complement System

Complement System

Complement System

- Serum Proteins
- Heat labile inactivated at 56°C for 30 minutes
- Lyse the bacterial cell
- Exist as Proenzymes which become activated after a cascade reaction

Complement System

Complement Functions of complement Proteins System

- Act as opsonin
- Chemoattractant for Polymorphonuclear Leucocytes (PMNs)
- Pro-Inflammatory: detrimental for host

Complement System

Complement System Activation

- Complement proteins are activated by following three pathways
- Classical Pathway
- Alternate Pathway
- Lectin Pathway

Immunology

Classical Pathway of Complement Activation

Classical Pathway of Complement Activation

Classical Pathway for Complement activation

- Activated by antibodies attached on pathogen (bacteria) Surface
- C1 protein is activated after interacting with Fc region of IgG or IgM
- C2 & C4 are activated after the C1 complex
- C3 is finally activated after the action of C2 & C4 in a cascade manner

End

Immunology

Alternate Pathway of Complement Activation

Alternate Pathway of Complement Activation

Alternate Pathway for Complement activation

- No need of antibody
- C3 is directly converted in the absence of antibodies
- Various co-factors of serum facilitate the activation of C3 sequentially
- Finally C3 helps in the lysis of bacterial cell

End

Immunology

Lectin Pathway of Complement Activation

Lectin Pathway of Complement Activation

Lectin Pathway for Complement activation

- Very similar to classical pathway
- In place of antibodies, mannose on the surface of specific pathogens are involved in activation

Lectin Pathway of Complement Activation

Lectin Pathway for Complement activation

- Mannose binding Lectins (MBL) in serum bind to surface of pathogen containing mannose in their cell wall e.g Fungus
- C3 proteins are produced after interaction of mannose & Lectins

Immunology

Effector Functions of Complement Proteins

Effector Functions of Complement Proteins

Effector Functions of Complement Proteins

- Opsonization (Opsonins)
- Chemoattraction (Chemoattractants)
- Anaphylaxis (Anaphylatoxins)
- Pro-Inflammatory

End

Immunology

Pathology related to Complement Proteins

Pathology Related to Complement Proteins

Pathology related to Complement system

- Acute Inflammatory response
- Capillary dilatation
- Exudation of plasma proteins & fluids (Edema)
- Bronchoconstriction
- Mast cells degranulation (Allergic reaction)

End

Immunology

Cells & Organs of the Immune System

Immunology

Tissues of Immune System

Tissues of Immune System

Tissues of Immune System

- Two (02) major types of tissues of immune system based on their functions
- 1. Primary (Generative) or Central Lymphoid Organs
- 2. Secondary or Peripheral Lymphoid Organs

Immunology

Peripheral Lymphoid Organs

Peripheral Lymphoid Organs

Primary Lymphoid Organs

- Generative or Central Lymphoid Organs
- Involved in maturation of Lymphocytes
- Contain Stem cells for division and maturation
- Bone Marrow
- Thymus

Immunology

Secondary Lymphoid Organs

Secondary Lymphoid Organs

Secondary Lymphoid Organs

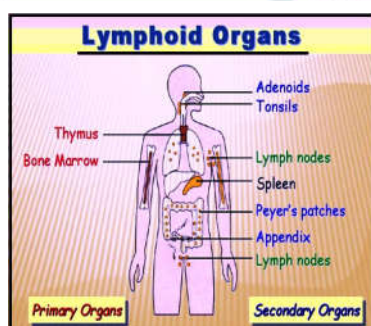
- Peripheral Lymphoid Organs
- Involved in providing adaptive immune response
- Site of interaction of antigen, antigen presenting cells & lymphocytes
- Various Anatomical Location

Secondary Lymphoid Organs

Secondary Lymphoid Organs

- Lymph Nodes
- Spleen
- Tonsils
- Adenoids
- Appendix
- Peyer's patches

Secondary Lymphoid Organs



Immunology

Secondary Lymphoid Organs (Merged by Talha Waraich

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Secondary Lymphoid Organs

Secondary Lymphoid Organs

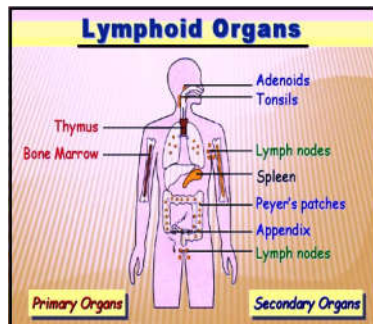
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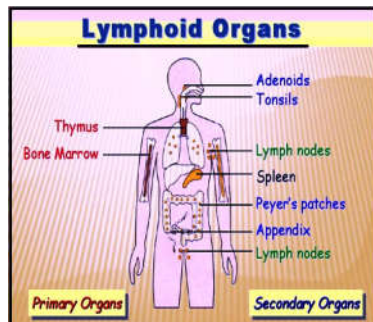


<http://exxamm.com/blog/Blog/14380/zxcvbnm4?>

Immunology

Secondary Lymphoid Organs

Secondary Lymphoid Organs



<http://exxamm.com/blog/Blog/14380/zxcvbnm4?>

Immunology

Lymphoid Lineage of Immune Cells

Lymphoid Lineage of Immune Cells

Lymphoid Lineage

- Lymphoid lineage progenitor cells give rise to the following immune cells
- B-Lymphocytes (Plasma cells or antibodies forming cells)
- T-Lymphocytes (T-Helper & Cytotoxic T-Cells)
- Natural Killer (NK) cells

Immunology

Lymphoid Lineage of Immune Cells

Lymphoid Lineage of Immune Cells

Lymphoid Lineage

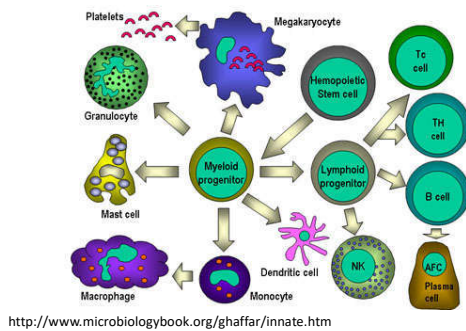
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Myeloid Lineage of Immune Cells

Myeloid Lineage

- Myeloid lineage progenitor cells give rise to the following immune cells
- Monocytes & Macrophages
- Dendritic cells
- Megakaryocytes
- Granulocytes

Cells & Organs of the immune system



Immunology

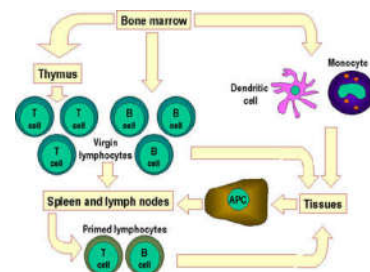
Lymphocyte Recirculation

Lymphocyte Recirculation

Lymphocyte Recirculation

- Naïve (Virgin) lymphocytes from primary lymphoid tissues move to secondary tissues
- Priming of lymphocytes
- Lymphocytes in blood enter in lymph node and percolate
- 1-2% of lymphocytes recirculate every hour

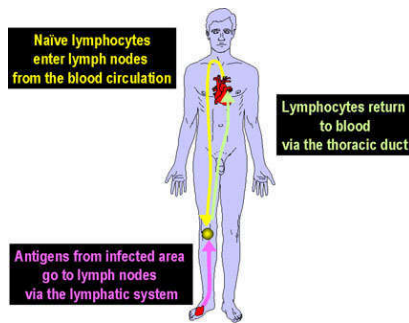
Lymphocyte Recirculation



Lymphocyte Migration & Localization

- Chances of successful encounter of specific B & T lymphocytes with appropriate antigen is low
- By lymphocytes migration, chances of successful encounter may increase
- Lymphocytes from blood enter in lymph nodes and percolate through lymph nodes

Lymphocyte Migration & Localization



Immunology

Clonal Selection of Lymphocyte

Clonal Selection of Lymphocyte

- T-Cell Receptors (TCR)
- Receptor Specificity
- Lymphocyte Repertoire
- After maturation, selection of lymphocytes which are non-reactive to self antigens

Clonal Selection of Lymphocyte

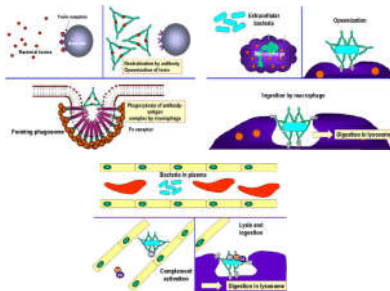
- Four principles
 - 1) Unique Receptor Specificity
 - 2) Lymphocyte Activation: Interaction with foreign molecule
 - 3) Differentiation of lymphocyte: from single clone
 - 4) Selection of lymphocytes bearing receptors against non-self molecules

Control of Extracellular Pathogens

Control of Extracellular Pathogens

- Secretory immune molecules are effective against extracellular pathogens like antibodies & complement proteins
- Three ways of controlling
 - 1) Neutralization
 - 2) Opsonization
 - 3) Complement activation

Control of Extracellular Pathogens



<http://www.microbiologybook.org/bowers/bact-comp.jpg>

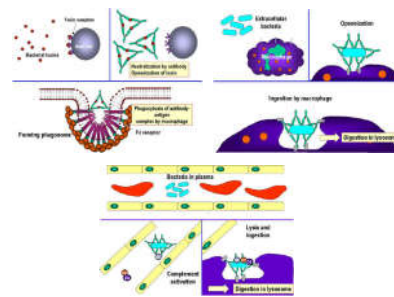
Immunology

Control of Extracellular Pathogens

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Control of Extracellular Pathogens



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Immunology

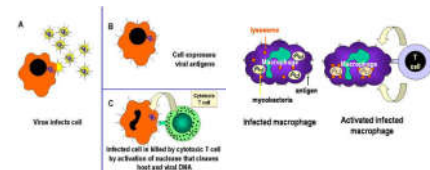
Control of Intracellular Pathogens
(Merged by
Talha Waraich
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Telegram at
03488485459)

Control of Intracellular Pathogens

- Secretory immune molecules are ineffective against intracellular pathogens like viruses & intracellular bacteria
- Cell mediated immune response is the primary defense against intracellular pathogens
- T-Lymphocytes play role in cell mediated immunity

Control of Intracellular Pathogens

- Cell mediated immune response varies according to the residing site of the pathogen
 - 1) Cytosolic site- Cytotoxic T-Cells e.g Viruses
 - 2) Vesicular site- Helper T-cells e.g *Mycobacterium tuberculosis*



Control of Cytosolic Pathogens

Control of Vesicular Pathogens

<http://www.microbiologybook.org/bowers/myco-macro2.jpg>

Properties of antibodies and antigens together with their structure, functions & interactions

Comparison B/W Antigen, Immunogen & Hapten

Comparison B/W Antigen, Immunogen & Hapten

- **Antigen**

A substance that reacts with the products of specific immune response

- **Immunogen**

A substance that induces specific immune response

- **Hapten**

A substance that is non-immunogenic but can react with products of specific immune response

Comparison B/W Antigen, Immunogen & Hapten

- **Hapten/Carrier**

Hapten has the property of antigenicity but in combination with carrier molecule it become immunogen

- **Epitope/Antigenic determinant**

The portion of antigen that combines with products of immune system

Factors influencing Immunogenicity (Contribution of the Immunogen)

Contribution of the Immunogen

- **Contribution of the Immunogen**

Following factors contribute in immunogenicity of Immunogen

- 1) **Foreignness**

Non-Self (Foreign) molecules induce immune response

- 2) **Chemical Composition**

Complex structured are more Immunogenic

Contribution of the Immunogen

- **Contribution of the Immunogen**

- 3) **Physical Form**

Particulate antigens: More immunogenic

Soluble antigens: Less immunogenic

- 4) **Degradability**

Easily degradable antigens are more immunogenic

Factors influencing Immunogenicity (Contribution of the Biological System)

Contribution of the Biological System

1) Genetic factors

- Some substances are immunogenic in some species but not in others
- Responders vs Non-Responders
- Altered genes which encode for B & T cells receptors

2) Age

Very young & old individuals have diminished immune response

Immunology

Factors influencing Immunogenicity (Method of Administration) (Merged by Talha Waraich

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Method of Administration

Administration of Immunogen

1) Dose

- Optimal dose of immunogen causes immune response
- Above or below of optimal dose remain insufficient for appropriate immune response

Method of Administration

Administration of Immunogen

2) Route

- Route of immunogen administration alter the nature of immune response
- Subcutaneous route: more immunogenic
- Intravenous & Gastric route: less immunogenic

Method of Administration

Administration of Immunogen

3) Adjuvants

- Substances which increase the immune response of an immunogen
- Used with vaccine in vaccination
- Alum or Aluminium hydroxide
- Show undesirable side effects i.e fever

Immunology

Chemical Nature of Immunogen

Chemical Nature of Immunogen

Chemical nature of Immunogen

1) Proteins

- Good Immunogens
- Most immunogens
- Pure & Conjugated

2) Polysaccharides

- Pure & Lipopolysaccharides are good immunogens

Chemical Nature of Immunogen

Chemical nature of Immunogen

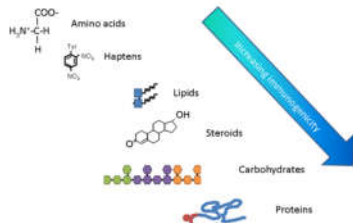
3) Nucleic Acids

- Poor Immunogens
- Immunogenic in single stranded form
- Complex with proteins

4) Lipids

- Non-Immunogenic
- Can be haptens

Immunogenicity



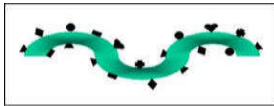
<http://www.microbiologybook.org/mayer/immunogenicity.gif>

Types of Antigens (T-Cell dependent & Independent)

Types of Antigens

T-dependent antigens

- Require T-cells
- B-cells are activated with the help of T-cells for antibodies production
- Protein in nature
- Contain variety of epitopes with few copies

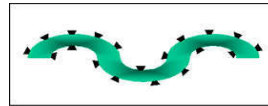


<http://www.microbiologybook.org/mayer/ag-2a.jpg>

Types of Antigens

T-independent antigens

- Not Require T-cells
- B-cells are activated directly without the help of T-cells for antibodies production
- Polysacchride in nature
- Contain same kind of epitopes in polymeric form
- More resistant to degradation



<http://www.microbiologybook.org/mayer/ag-1.jpg>

Superantigens

Superantigens

Superantigens

• Definition	Conventional Antigen	Superantigen
	Monovalent/Oligoclonal response $1:10^4 - 1:10^5$	Polyclonal response $1:4 - 1:16$

<http://www.microbiologybook.org/mayer/antigens2000.htm>

Superantigens

- T-dependent antigens
- Conventionally activate small fraction of T-cells
- Superantigens can activate 25% of T-cells polyclonally
- Hyperactivation of T-cells by superantigens
- Bacterial antigens e.g Staphylococcal enterotoxin (Food Poisoning)

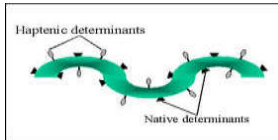
Superantigens

Superantigens

- Bacterial antigens e.g Staphylococcal enterotoxin (Food Poisoning), Staphylococcal Exfoliatin toxin (Scalded Skin Syndrome), Staphylococcal Toxic shock toxin (Toxic Shock Syndrome)
- Viral & other microorganisms antigens can also be superantigens

Hapten-Carrier Conjugate

Hapten-Carrier Conjugate



<http://www.microbiologybook.org/mayer/agg-3.jpg>

- Immunogenic molecules to which hapten is non-immunogenic
- Carrier is immunogenic in nature
- Hapten bounded with carrier covalently
- Structurally contain Haptenic determinants & Carrier determinants
- Immune response is being determined by Carrier determinant

Immunology

Antigenic determinants recognized by Innate Immune System

Antigenic determinants/Innate Immune

- No specificity
- Broad Molecular pattern called Pathogen associated molecular patterns (PAMPs) present on different pathogens
- PAMPs are recognized by Pattern recognition Receptors (PRRs) on the surface of immune cells

Antigenic determinants/Innate Immune

PAMPs	PRRs	Effector Function
LPS	Toll Like Receptors-4 (TLR-4)	Macrophage activation
Flagellin	TLR-5	Macrophage activation
Microbial cell wall components	Complement	Opsonization, Complement activation

Immunology

Characteristics of Immunoglobulins (Antibodies)

Characteristics of Immunoglobulins

Immunoglobulins

- Glycoproteins
- Produced by plasma cells in response to immunogens
- Globular proteins responsible for immunity (Humoral)
- Has ability to combine specifically with antigens

Characteristics of Immunoglobulins

Characteristics

Antigen (Ag) binding

- Primary function
- To bind specifically with epitopes of antigens
- Paratope: Binding site

Valency

- Number of antigenic determinants which can bind with immunoglobulin
- Minimum is two (02)

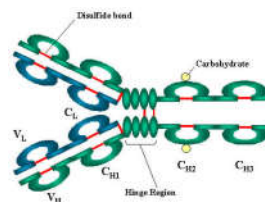
Characteristics of Immunoglobulins

Characteristics

Effector functions

- After binding with antigen, there is no primary direct biological function
- Secondary effector functions
 - 1) Complement activation
 - 2) Binding to various cell types via defined receptors (Fc receptors)

Basic Structure of Antibodies

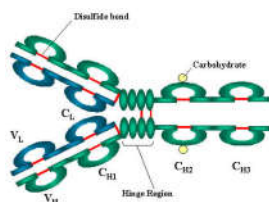


<http://www.microbiologybook.org/mayer/stru-2.jpg>

Immunoglobulins

- Composed of following structures
- 1) Heavy & Light chains: Four chains, 02 identical light chains, 02 identical heavy chains
- 2) Disulfide bonds: Interchain & Intrachain
- 3) Variable & constant regions: based on amino acid sequences

Basic structure of Antibodies



<http://www.microbiologybook.org/mayer/stru-2.jpg>

Immunoglobulins

- Hinge region: makes antibodies flexible to change its shape while performing function
- Domains: Folded regions which contain an intra-chain disulfide bond
- Oligosaccharides: attached on CH2 domain in most of immunoglobulins

Immunology

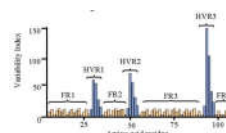
Structure of the Variable Region

Structure of the Variable Region

Variable region

- Present on heavy & light chains of Immunoglobulins
- Two sub-regions
- 1) Hypervariable (HVR)
 - Based on variability of amino acid sequences
 - Complementarity determining region

Structure of the Variable Region



<http://www.microbiologybook.org/mayer/stru-3.jpg>

Variable Region

- Framework Regions
 - Regions between hypervariable regions
 - Groups & sub-groups of immunoglobulins
 - Products of different variable region genes

Structure of the Variable Region

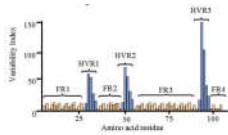
(Merged by
Talha Waraich
For help What's App or
Telegram at
03488485459)

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Structure of the Variable Region



<http://www.microbiologybook.org/mayer/stru-3.jpg>

Variable Region

2) Framework Regions

- Regions between hypervariable regions
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- Products of different variable region genes

Immunology

Antibodies Fragments Structure/Function

Antibodies Fragments

Antibodies Fragments

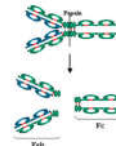
- Produced by proteolytic enzymes i.e papain
- Two fragments are produced which have structure & function relationship

- 1) Fab
- 2) Fc

Antibodies Fragments

Immunoglobulin Fragments: Structure/Function Relationships

- Fab
 - Ag binding
 - Valence = 1
 - Specificity determined by V_H and V_L
- Fc
 - Effector functions



<http://www.microbiologybook.org/mayer/stru-3.jpg>

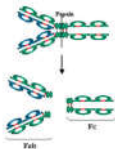
1) Fab

- Fragment for antigen binding (Fab)
- Papain causes the cleavage at hinge regions to produce two identical fragments containing one heavy & one light chain
- Having monovalent valency for antigen binding

Antibodies Fragments

Immunoglobulin Fragments: Structure/Function Relationships

- Fab
 - Ag binding
 - Valence = 1
 - Specificity determined by V_H and V_L
- Fc
 - Effector functions



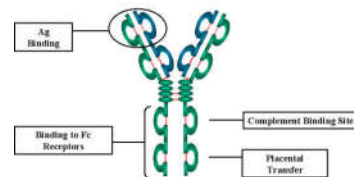
<http://www.microbiologybook.org/mayer/stru-3.jpg>

2) Fc

- Crystallizable Fragment (Fc)
- Contain two heavy chains with CH2 & CH3 domains
- Performed effector functions

Antibodies Fragments

Immunoglobulin Fragments: Structure/Function Relationships



<http://www.microbiologybook.org/mayer/stru-5.jpg>

Immunology

Classes of Antibodies

(Merged by Talha Waraich

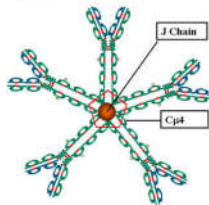
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Immunology

IgM/IgG Properties & Functions

IgM & IgG Properties & Functions

IgM



<http://www.microbiologybook.org/mayer/stru-8.jpg>

IgM

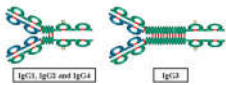
- Carry Mu heavy chain
- Structurally, IgM exist in pentamer generally as secretory form
- Monomer form also exist but membrane bounded on B-Cells
- All five molecules are joined through J-chain

IgM & IgG Properties & Functions

IgM

- Third most common serum antibody
- First antibody made by virgin B-cells
- Due to structure, IgM is a good complement fixing antibody
- Bind to surface of microorganisms for removal by phagocytosis

IgM & IgG Properties & Functions



<http://www.microbiologybook.org/mayer/stru-7.jpg>

IgG

- Carry Gamma heavy chain
- Structurally, IgG exist in monomer
- Has subclasses like IgG1, IgG2, IgG3 & IgG4 based on number of disulphide bonds & length of hinge region

IgM & IgG Properties & Functions

IgG

- Major antibody of serum-75% of total antibodies
- Also major in extravascular space
- Complement fixing
- Role in placental transfer: by FcR on placental cells
- Good opsonin

Classes of Antibodies

Classes

- Classes of antibodies: based on amino acid sequences in the constant region of heavy chains
- Five different heavy chains on which classes of antibodies are based

Classes of Antibodies

Classes

- 1) Gamma chain- IgG
- 2) Mu chain- IgM
- 3) Alpha chain-IgA
- 4) Delta chain-IgD
- 5) Epsilon chain-IgE

Immunology

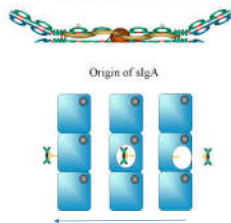
IgA/IgE/IgD Properties & Functions

IgA/IgE/IgD Properties & Functions

IgA

- Second most common serum antibody
- In Serum, IgA exist as monomer
- Mostly IgA is present in secretions i.e saliva, tears, mucus & colostrum
- Responsible for "Mucosal Immunity"

IgA/IgE/IgD Properties & Functions



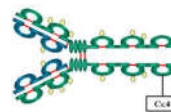
<http://www.microbiologybook.org/mayer/stru-12.jpg>

IgA

- In secretions. two molecules of IgA are linked with J chain
- sIgA contains secretory piece
- Secretory piece is added to IgA in epithelial cells
- IgA doesn't fix complement unless aggregated

IgA/IgE/IgD Properties & Functions

IgE

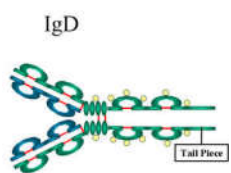


<http://www.microbiologybook.org/mayer/stru-14.jpg>

IgE

- Least common serum antibody
- IgE exist as monomer
- Involved in allergic reactions (Anaphylaxis)
- Degranulation of basophils after IgE binding
- Responsible for parasite killing: IgE mediated
- Not fix complement

IgA/IgE/IgD Properties & Functions



<http://www.microbiologybook.org/mayer/stru-13.jpg>

IgD

- Very low level serum antibody
- Role in serum is unknown
- Primarily found on B-cells as receptor for antigen with help of tail piece
- Exist as monomer
- Doesn't fix complement

Immunology

Types vs Sub-types of Antibodies

Types vs Sub-Types of Antibodies

Types

- Based on kind of light chains of antibodies
- Following two light chains
 - 1) Kappa light chains
 - 2) Lambda light chains
- Based on differences in amino acid sequences in constant regions of light chains e.g IgG Kappa, IgG Lambda

Sub-Types

- Light chains are divided into subtypes based on differences in amino acid sequences in constant region
- Following Subtypes of light chains
- Lambda chains: 4 subtypes e.g L1, L2, L3 & L4
- Kappa chains: 2 subtypes e.g K1 & K2

Immunology

Antigen & Antibody Reactions

Antigen & Antibody Reactions

Nature of Ag & Ab reactions

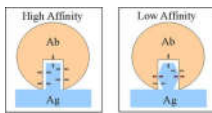
- Interactions B/W antigen & antibody (Epitope & Paratope)
- Serological reactions
- One is known & one is unknown
- Lock & Key Concept

Nature of Ag & Ab reactions

- Non-Covalent bonding
- Hydrogen bonds
- Vander Wall forces
- Electrostatic forces
- Reversibility

Factors Effecting on Antigen & Antibody Tests

Factors effecting on Ag & Ab Tests



<http://www.microbiologybook.org/mayer/rx-2.jpg>

Factors

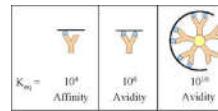
- Easily detectable Ag & Ab reactions

1) Affinity

- Strength of binding B/W single antigenic determinant & single antibody site
- Sum of attractive & repulsive forces
- Higher the affinity of Ab for Ag, more stable interaction

Factors effecting on Ag& Ab Tests

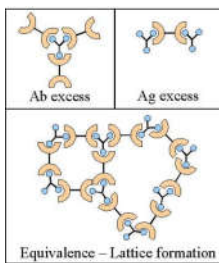
2) Avidity



<http://www.microbiologybook.org/mayer/rx-4.jpg>

- Strength of binding of antigen with many antigenic determinant & multivalent antibody
- Overall strength B/W multivalent antigens & antibodies
- Avidity is influenced by valency of both Ag & Ab
- More avidity more stable & easy to detect

Factors effecting on Ag& Ab Tests



<http://www.microbiologybook.org/mayer/rx-6.jpg>

3) Ag & Ab ratio

- Ag & Ab ratio determines the nature of complex
- Lattice formation: Equivalence B/W Ag & Ab ratio
- Larger Ag & Ab complex: Easily detectable

Factors effecting on Ag& Ab Tests

4) Physical form

- Physical form of antigen determines the nature of Ag & Ab reaction
- Particulate form of Ag: Agglutination reactions
- Soluble form of Ag: Precipitation reactions

Agglutination Reactions

Agglutination

- Particulate nature of antigen causes the reaction of antibody to form visible clumps (Agglutinate).
 - Antibodies: Agglutinins
 - Nature of particle
- 1) RBC: Hemagglutination
 - 2) Bacteria: Bacterial agglutination
 - 3) Latex: Latex Agglutination

Agglutination Reactions



<http://www.microbiologybook.org/mayer/rx-7.jpg>

Hemagglutination

- Qualitative: Blood group typing (ABO & Rh)
- Quantitative: For determining the quantity of antibodies
- Titer: the lowest conc. Of Ab which causes agglutination or the max. dilution of serum which causes agglutination

Agglutination Reactions

Applications of Agglutination

- Blood grouping: ABO & Rh for both either on the basis of antigens of RBC or antibodies of serum
- Assessing bacterial infections e.g Widal test for diagnosis of Typhoid fever

Immunology

Precipitation Reactions

Precipitation Reactions

Precipitation (Diffusion)

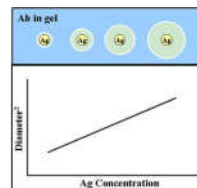
- Soluble nature of antigen causes the reaction of antibody to form precipitate in a medium
- Immunoprecipitation
- Antibodies: Precipitins
- Medium used for visualization of Ag & Ab reaction e.g Agar
- Both Ag & Ab are soluble form in serum

Precipitation Reactions

Precipitation

- On the basis of diffusion of each molecule i.e Ag & Ab, two types of precipitation reactions
- 1) Single immunodiffusion or precipitation
 - 2) Double immunodiffusion or precipitation

Precipitation Reactions



<http://www.microbiologybook.org/mayer/rx-13.jpg>

Single Immunodiffusion

- Also called Radial Immunodiffusion
- Antibody is already incorporated in agar
- Serum containing antigen is added
- Antigen diffuses in agar & react with Ab to form visible precipitate
- Used for determining the level of serum immunoglobulins e.g IgA

Precipitation Reactions

Double Immunodiffusion

- Both antigen & antibody is added in the medium separately
- Ag & Ab are allowed to react together and reached at equivalence to form precipitate
- Qualitative analysis of complex mixture of Ag can be determined
- Purity of isolated serum proteins can be checked

Immunology

Complement Fixation Tests

Complement Fixation Test

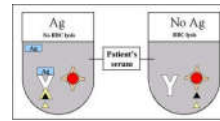
Complement Fixation

- Ag & Ab reaction which can be detected on the basis of ability to fix or consume complement
- Used for determining good complement fixing antibodies e.g IgG or IgM
- Unknown Ag : Known Ab
- Complement proteins
- Presensitized RBC with anti RBC antibodies

Complement Fixation

Complement Fixation Interpretation

- Lysis of RBC: No Complement fixation by Ag & Ab complex: No Ag: Negative CFT
- No Lysis: Complement fixation by AG & Ab complex: Ag is present: Positive CFT
- CFT is used for determining Gonococcal Ab in serum



<http://www.microbiologybook.org/mayer/rx-24.jpg>

Immunology

Enzyme Linked Immunosorbent Assay (ELISA)

ELISA

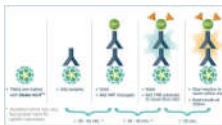
ELISA

- Ag & Ab reaction which can be detected on the basis of substrate utilization by enzyme
 - Solid phase Immunoassay
 - Used for determining either unknown Ag or Ab
- 1) Direct ELISA
 - 2) Indirect or Sandwich ELISA

ELISA

Direct ELISA

- Unknown Ab can be determined
- Known Ag is coated on solid phase (Glass surface)
- Serum containing unknown Ab is added
- Another ab which is precoated with enzyme e.g horseradish peroxidase
- Substrate: H_2O_2

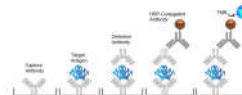


<https://www.stellarbiotechnologies.com/products/order-stellar-klh-elisa-kits>

ELISA

Indirect (Sandwich) ELISA

- Unknown Ag can be determined
- Known Ab is coated on solid phase (Glass surface)
- Serum containing unknown Ag is added
- Another Ab which is precoated with enzyme e.g horseradish peroxidase
- Substrate: H_2O_2



<https://www.lsbio.com/elisakits/human-f7-factor-vii-elisa-kit-sandwich-elisa-ls-f10416/10416>

ELISA

Applications

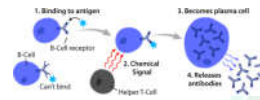
- Used for determining viral antigens in patient's serum e.g Hepatitis B surface Antigens (HbsAG)
- For determination of antibodies against viruses e.g Anti HCV antibodies
- For serum cytokine levels e.g IL-1, TNF-a etc

Immunology

Genetics of Antibody Structure & Diversity

Antibody Formation

Antibody Formation



<https://www.quora.com/What-are-the-differences-between-B-Cells-and-T-Cells>

- A process of immune response against antigen or immunogen
- Can be T-independent or dependent
- B-cells recognize the Ag with its specific receptor
- Transformation of B-cells into plasma cells
- Plasma cells secrete antibodies

General Characteristics of Antibody Response

General Characteristics of Antibody Formation

General Characteristics

- Antibody formation process is well controlled mechanisms with following characteristics
- 1) Self/Non-Self Discrimination
- Antibodies would be reactive against non-self antigens
- Reactivity against self Ag: Autoimmunity

General Characteristics of Antibody Formation

General Characteristics

- 2) Memory
- Remembering the nature of antigen
- Illicit memory response against same antigen
- Anamnestic response
- Robust way of clearing antigen
- B memory cells

General Characteristics of Antibody Formation

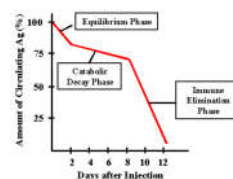
General Characteristics

- 3) Specificity
- High degree of specificity of antibodies against specific antigens
- Less chances of cross reactivity
- Characteristic of adaptive immune response

Fate of Immunogen

Fate of Immunogen

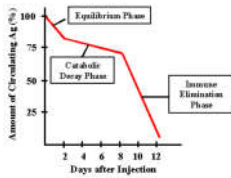
Fate of Immunogen Primary Injection



<http://www.microbiologybook.org/mayer/ab1-1.jpg>

- Immunogen enters in the body first time (Primary Injection)
- Following four phases for kinetics of immunogen clearance
- 1) Equilibrium Phase
- Rapid diffusion process
- As equilibrates B/W vascular & extravascular components by diffusion

Fate of Immunogen

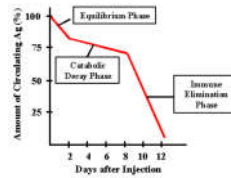


<http://www.microbiologybook.org/mayer/ab1-1.jpg>

Fate of Immunogen Primary Injection

- 2) Catabolic decay phase
 - Host immune cells cause the decay of Ag
 - Cells & enzymes involved in this phase
 - Macrophages & phagocytic cells
 - Duration depends upon the nature of immunogen & host

Fate of Immunogen



<http://www.microbiologybook.org/mayer/ab1-1.jpg>

Fate of Immunogen Primary Injection

- 3) Immune Elimination phase
 - Newly synthesized antibodies combine with Ag to form Ag/Ab (Immune) complexes
 - Immune complexes: eliminated by phagocytosis
 - Ab appears in serum only after this phase gets over

Fate of Immunogen

Fate of Immunogen Secondary Injection

- If there is circulating Ab: rapid immune elimination of Ag
- No circulating Ab: All three phases occur but immune elimination phase will be rapid

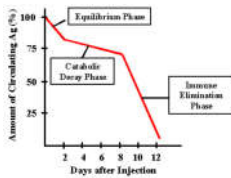
Immunology

Fate of Immunogen

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Fate of Immunogen

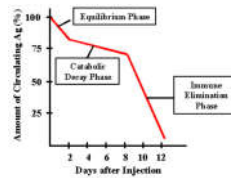


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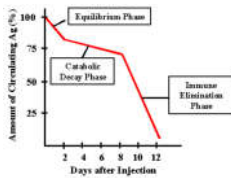


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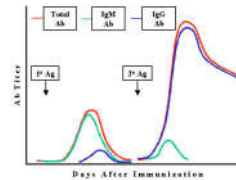
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Primary & Secondary Antibody Response



<http://www.microbiologybook.org/mayer/ab1-4a.jpg>

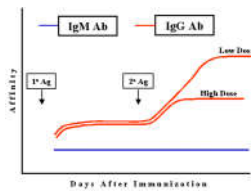
Primary Ab response

- First kind of antibody response against antigen
- Major class of antibody is IgM
- Ab are with less affinity & Less persistent

Secondary Ab response

- Second kind of antibody response against antigen
- Major class is IgG but may IgE or IgA but less IgM

Primary & Secondary Ab Response



<http://www.microbiologybook.org/mayer/ab1-5.jpg>

Primary Ab response

- Ab are with less affinity

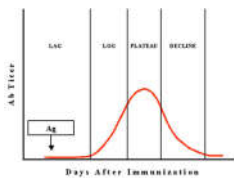
Secondary Ab response

- Ab with high affinity
- With low dose of Ag, Ab have high affinity as compare to high dose (Affinity maturation)

Immunology

Kinetics of antibody response against T-dependent antigens

Kinetics of Ab/T-dependent Ag

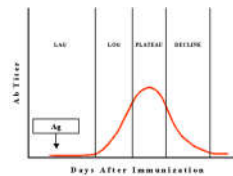


<http://www.microbiologybook.org/mayer/ab1-2.jpg>

Primary Ab response

- Four phases
- 1) Inductive or Lag phase
 - Antigen is recognized as foreign
 - Immune cells start to proliferate & differentiate in response to antigen
 - Duration: Usually 5-7 days

Kinetics of Ab/T-dependent Ag



<http://www.microbiologybook.org/mayer/ab1-2.jpg>

Primary Ab response

- 2) Log or Exponential phase
 - Ab conc. Increase rapidly
 - B-cells proliferation
 - B-cells differentiation into plasma cells to secrete more antibodies

Kinetics of Ab/T-dependent Ag

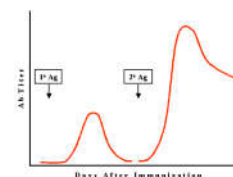


<http://www.microbiologybook.org/mayer/ab1-2.jpg>

Primary Ab response

- 3) Plateau or Steady State phase
 - Ab synthesis is balanced by Ab decay
 - No net increase in Ab concentration
- 4) Decline or Decay Phase
 - Rate of Ab decay exceeds
 - Fall in Ab level
 - Base line Ab level

Kinetics of Ab/T-dependent Ag

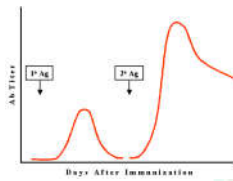


<http://www.microbiologybook.org/mayer/ab1-3.jpg>

Secondary Ab response

- Second time (Anamnestic) exposure would also induce the same four phases but with different kinetics
- 1) Lag phase
 - Shorter as compare to primary exposure
- 2) Log phase
 - More rapid & sharp
 - Higher levels of Ab in short time

Kinetics of Ab/T-dependent Ag



<http://www.microbiologybook.org/mayer/ab1-3.jpg>

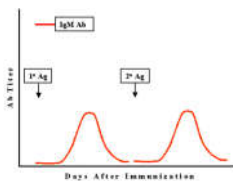
Secondary Ab response

- 3) Plateau or Steady state phase
 - No plateau
- 4) Decline phase
 - Less rapid & sharp
 - Antibodies may persist for days, months or even years due to memory response

Immunology

Kinetics of antibody response against T-independent antigens

Kinetics of Ab/T-independent Ag



<http://www.microbiologybook.org/mayer/ab1-9.jpg>

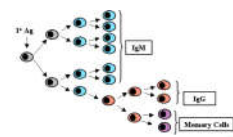
T-independent Ag

- Mainly IgM production
- All those phases in primary response
- No secondary response as same phases in second time exposure to Ag
- No memory response

Immunology

Cellular Events during Antibody Response

Cellular events during Ab Response



<http://www.microbiologybook.org/mayer/ab1-7.jpg>

Primary Response

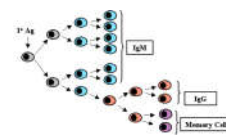
Lag phase

- Clones of B & T-cells bind to antigens with receptor
- B-Cells differentiate into plasma cells
- Plasma cells start to secrete Ab

Log Phase

Plasma cells initially secrete IgM as Mu chains gene is close to rearranged VDJ regions

Cellular events during Ab Response



<http://www.microbiologybook.org/mayer/ab1-7.jpg>

Primary Response

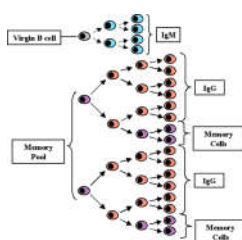
Plateau phase

- Due to Ag depletion & T-cells are no longer active
- Plasma cells begin to die
- Newly synthesized Ab equilibrates

Decline Phase

- No new Ab formation
- Most of plasma cells die

Cellular events during Ab Response



<http://www.microbiologybook.org/mayer/ab1-8.jpg>

Secondary Response

- Memory cells pool: Comprised of T & B cells activated during primary response
- Mostly memory pool cells are activated
- Ab class switching: IgM to IgG
- Mostly IgG in secondary response
- Some plasma cells differentiate into memory

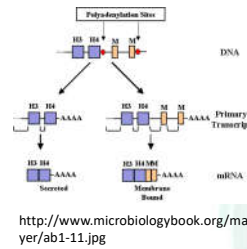
Immunology

Membrane & Secreted Antibodies

Membrane & Secreted Antibodies

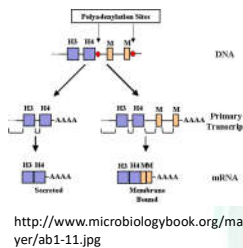
- Membrane bounded & secreted Ab from a same B-cell have same specificity
- Role of Immunoglobulin gene in determining the specificity
- Polyadenylation sites in gene for Ig determine the nature of immunoglobulin

Membrane & Secreted Antibodies



- There are two kinds of polyadenylation sites present in gene of Ab
- 1) After the exons of last heavy chain (H)
- 2) After the exons which encode for transmembrane domain (M)
- If the first sites is used the pre-mRNA processed for secretory form of immunoglobulin

Membrane & Secreted Antibodies



- If the second polyadenylation site is used the pre-mRNA processed for membrane bounded form of immunoglobulin
- As the gene contain same VDJ region that's why specificity remain same

Immunology

Expression of Immunoglobulin Genes & V(D)J Recombination

Immunology

Structure of Human Antibody Gene (Loci)

Structure of Human Antibody Gene

Antibody Gene

- Gene: Protein
- Immunoglobulin repertoire is encoded by multiple germ line gene segments
- These segments undergo somatic recombination during development of B-Cells
- Basic component of gene is inherited but alteration during lifetime

Structure of Human Antibody Gene

Light Chain Gene Family

Located at Chromosome 22

Lambda light chain genes; n=30



http://www.microbiologybook.org/mayer/gen-1.jpg

- 1) Lambda Light Chains
- Composed of 4 C region genes. One for each subtype of Lambda
 - 30 Variable genes
 - Each of V gene composed of L (Ladder) exons
 - V (variable) Exons
 - J (Joining) exons
 - Introns in between Exons

Structure of Human Antibody Gene

Light Chain Gene Family

Located at Chromosome 2

Kappa light chain genes; n=300



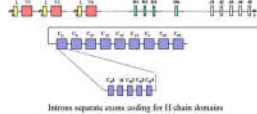
http://www.microbiologybook.org/mayer/gen-1.jpg

- 2) Kappa Light Chains
- Composed of only single C region gene for a single type of Kappa
 - 300 Variable genes
 - Each of V gene composed of L (Ladder) exons
 - V (variable) Exons
 - J (Joining) exons
 - Introns in between Exons

Structure of Human Antibody Gene

Located at Chromosome 14

Heavy chain genes, V_H*1000, D_H*15



<http://www.microbiologybook.org/mayer/gen-3.jpg>

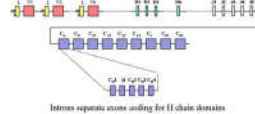
Heavy Chain Gene Family

- Located at chromosome 14
- Composed of many C region gene for a each class & sub class
- Each of C contain exons for hinge region & domains
- 1000 Variable genes

Structure of Human Antibody Gene

Located at Chromosome 14

Heavy chain genes, V_H*1000, D_H*15



<http://www.microbiologybook.org/mayer/gen-3.jpg>

Heavy Chain Gene Family

- Each of V gene composed of L (Ladder) exons
- V (variable) Exons
- Additional D (diversity) Exons
- J (Joining) exons
- Introns in between Exons

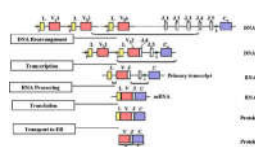
Immunology

Somatic Recombination
(Merged by
Talha Waraich
For help What's
App or Telegram at
03488485459)

Somatic Recombination

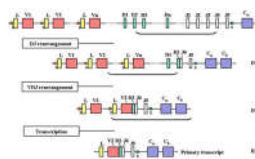
- Recombination involves rearrangement of DNA in somatic cells
- The newly recombined genes are not inherited in contrast to germ cells
- Primary Ig repertoire differ slightly from one individual to next one
- Also differ in individual's lifetime by their exposure to different antigens

Somatic Recombination



<http://www.microbiologybook.org/mayer/gen-2.jpg>

- As B-cell differentiate into mature one, it make light chain (Kappa)
- There is rearrangement of genes(exons) as introns are removed and genes start to express
- In this recombination, V genes become close to J genes by removing introns
- Recombined DNA processed into mature RNA by splicing



<http://www.microbiologybook.org/mayer/gen-4.jpg>

- As B-cell differentiate into mature one, it make heavy chains
- There is rearrangement of genes(exons) as introns are removed and genes start to express
- In this recombination, D genes become close to J & then V recombines with DJ
- Recombined DNA processed into mature RNA by splicing

Immunology

Somatic Hypermutation

Somatic Hypermutation

- The enhanced rate of point mutation in the Ig V region genes
- Somatic hypermutation occurs particularly in V gene which codes for 2nd hypervariable region
- Increase in Ig diversity after various antigenic stimulation
- Also increase the affinity of Ig in order to compete for limited amount of Ag present

Role of Somatic Hypermutation in diversity of antibodies

Antibody Diversity

- Sum of all the possible Ab specificities that an organism can make
- Humans can make 10^7 - 10^8 different Ab molecules
- Increase in Ig diversity after various antigenic stimulation
- So increase in Ab specificities due to somatic mutation

Role of Somatic Hypermutation

Antibody Diversity

- Somatic hypermutation occurs at high rate approx. 10^6 times higher
- The exact mechanism by which mutation occurs in V region gene without effecting the C region is still under research
- Activation induced cytidine deaminase (AID): essential role in DNA deamination

Immunology

V(D)J combinational Diversity

V(D)J Combinational Diversity

Combinational Diversity

- The component of Ab diversity that is generated by joining of various gene segments
- In Light chain genes: V & J region genes combination
- In Heavy Chain: V, D & J region genes combination
- D combines first J and then V to form heavy chain

Immunology

V(D)J combinational Diversity

V(D)J Combinational Diversity

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Immunology

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V(D)J Combinational Diversity

Combinational Diversity

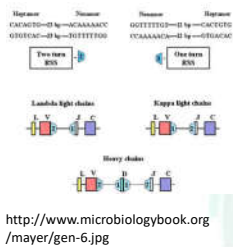
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Immunology

Regulation of V(D)J Recombination

Regulation of V(D)J Recombination

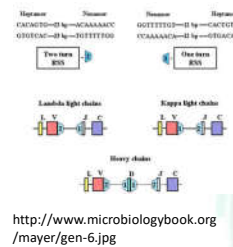
Regulation



- Recombination Signal Sequence (RSS): Flanked B/W V, D & J exons
- Composed of conserved heptamer & nonamer separated by 2 & 1 turn signals
- Recombination occurs B/W 2 & 1 turn

Regulation of V(D)J Recombination

Regulation



- Recombination occurs after the removal of introns B/W V & J in case of light chain
- Recombination occurs after the removal of introns B/W V, D & J in case of heavy chain
- RAG1 & RAG2: Enzymes responsible for this recombination
- SCID

Immunology

Affinity Maturation

Affinity Maturation

Affinity Maturation

- Increase in average affinity of Immunoglobulins
- Following Ag activation, V regions of heavy & light chains are diversified by somatic hypermutation
- This results in increase in binding affinity of BCR for its cognate ligand

Affinity Maturation

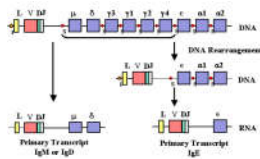
Affinity Maturation

- B-cells with higher affinity Ig can compete better for limited amount of Ag
- Increase in binding strength of Ab produced by immune response with Ag
- Affinity maturation occurs during secondary Ab response

Immunology

Class Switching or Class Switch Recombination

Class Switching

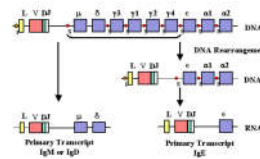


<http://www.microbiologybook.org/mayer/ab1-10.jpg>

Class Switching

- The process by which a B-Cell changes the class of Ab but not specificity
- Switching occurs from IgM to IgG or IgA & IgE
- Class switching occurs as a result of recombination at gene of Immunoglobulin
- Class switch recombination (CSR)

Class Switching



<http://www.microbiologybook.org/mayer/ab1-10.jpg>

Class Switching

- Recombination occurs B/W the rearranged VDJ & the switch site called S μ before the C μ (μ)
- This recombination causes to bring VDJ region to constant regions of other heavy chains except delta
- CSR occurs due to cytokines secreted by activated T-cells

Immunology

Antigen Processing & Presentation

Immunology

Review of B & T-Cell Receptors for Antigens

Review of B&T-cell Receptors

B-Cell Receptors (BCR)

- B & T cells recognize various form of substances as antigens in different & unique ways
- The role of receptors is very significant in such form of antigen recognition
- BCR is composed of surface immunoglobulin molecule with a certain specificity for antigen

Review of B&T-cell Receptors

B-Cell Receptors (BCR)

- BCR can recognize following antigens in soluble form
- Proteins (Both Native & Denatured determinants)
- Nucleic acids
- Polysaccharides
- Some lipids
- Small molecules i.e. haptens

Review of B&T-cell Receptors

T-Cell Receptors (TCR)

- TCR recognize mainly those antigens which are protein in nature
- Antigens in fragmented form (Processed)
- Not soluble form
- Processed proteins in association with Major Histocompatibility Complex (MHC)
- MHC expressed on all nucleated cells

Review of B&T-cell Receptors

T-Cell Receptors (TCR)

- T-cells are grouped functionally on the basis of associated MHC with fragmented protein fragments
- Cytotoxic T-Cells- MHC I
- Helper T-Cells- MHC II

Introduction to Antigen Processing & Presentation (Merged by Talha Waraich For help What's App or Telegram at 03488485459)

Ag Processing

- Processes that occur within the cell for fragmentation of antigens (Proteolysis)
- Association of fragmented peptide with Major Histocompatibility Complex (MHC) molecule expressed on the surface of Ag presenting Cells (APC) e.g Macrophages

Antigen Processing & Presentation

Ag Processing

- Presentation of processed Antigen with MHC molecule to T-cell determine the function of T-cell too
- Cytotoxic T-Cells: MHC I
- Helper T-Cells- MHC II

Immunology

Antigen Processing & Presentation of Endogenous antigens

Antigen Processing of Endogenous Ag

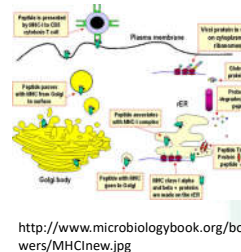
Endogenous Antigens

- Intracellular source of antigen: Endogenous Ag residing in cytosol e.g Viruses
- MHC I present such kind of Ag
- MHC I express on all nucleated cells
- Ag are processed in proteasome: a complex having proteolytic activity

Antigen Processing of Endogenous Ag

Endogenous Antigens

- Fragmented proteins move across the membrane of ER using transporter membrane
- Synthesis & assembly of MHC I complex inside ER
- Within ER MHC I form a stable complex with fragmented peptide & express on the surface of cell membrane

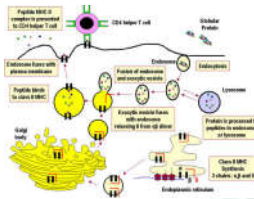


Antigen Processing & Presentation of Exogenous antigens

Antigen Processing of Exogenous Ag

Exogenous Antigens

- Exogenous antigens e.g bacteria are taken up by the process of endocytosis
- MHC II present such kind of Ag
- MHC I express on limited number of cells like antigen presenting cells (APC)
- APC include macrophages, dendritic cells & B-cells



<http://www.microbiologybook.org/bowers/fig2-mhc2.jpg>

Exogenous Antigens

- Exogenous proteins are processed in endosomes by proteases
- Synthesis & assembly of MHC II complex inside ER and transported across Golgi complex
- Trans GC combines with endosomes containing fragmented peptides
- MHC II complexed with peptides present on cell

Major Histocompatibility Complex (MHC)

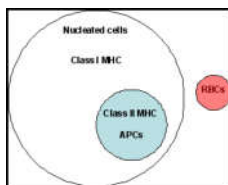
Introduction to MHC

Introduction to MHC

MHC

- Adaptive Immunity: Cell to cell interaction or Cell mediated immunity
- Cell to cell interaction orchestrated by immunological synapses
- TCR: T-cells
- MHC with processed peptides: on APC
- Two classes of MHC
- Class I & Class II

Introduction to MHC



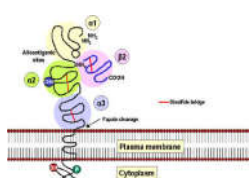
<http://www.microbiologybook.org/bowers/class-I-and-II-MHC.gif>

MHC

- MHC encoded by genes which are highly polymorphic
- MHC genes were first identified in case of tissue transplant rejection
- Class I MHC: on all nucleate cells of body
- Class II MHC: limited to APC

Structure of Class I MHC molecules

Structure of Class I MHC molecules

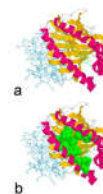


<http://www.microbiologybook.org/bowers/mhc1.jpg>

Class I MHC

- Composed of two polypeptide chains
- 1) A long alpha chain
- 2) A short beta chain
- A Cytoplasmic region
- A trans membrane region
- A highly conserved $\alpha 3$ region
- A highly polymorphic peptide binding region

Structure of Class I MHC molecules

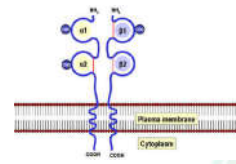


<http://www.microbiologybook.org/bowers/mhc1-pocket.gif>

Antigen binding Groove of Class I MHC

- Composed of $\alpha 1$ & $\alpha 2$ domain
- Antigenic peptide reside within the Ag binding groove
- Within groove peptide make contact with residue peptide (highly polymorphic)
- 8-10 amino acids can accommodate

Structure of Class II MHC molecules

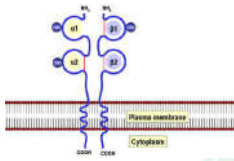


[http://www.microbiologybook.org/bo
wers/mhc2.jpg](http://www.microbiologybook.org/bo wers/mhc2.jpg)

Class II MHC

- Composed of two polypeptide chains α & β chains of equal length
- A Cytoplasmic region for phosphorylation & Binding to cytoskeleton
- A trans membrane region for anchoring the membrane

Structure of Class II MHC molecules

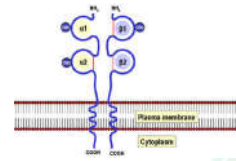


[http://www.microbiologybook.org/bo
wers/mhc2.jpg](http://www.microbiologybook.org/bo wers/mhc2.jpg)

Class II MHC

- A highly conserved $\alpha 2$ & $\beta 2$ domain for binding to CD4
- A highly polymorphic peptide binding region formed of $\alpha 1$ & $\beta 1$
- Ag peptide accommodates in Ag binding groove

Structure of Class II MHC molecules



[http://www.microbiologybook.org/bo
wers/mhc2.jpg](http://www.microbiologybook.org/bo wers/mhc2.jpg)

Class II MHC

- Peptide also remain in contact with residues which are highly polymorphic
- As groove is open it can accommodate approx. 13-25 amino acids long
- Can bind different amino acids due to its polymorphic nature

Important Aspects of MHC

Important Aspects of MHC

Class II MHC

- MHC molecules are membrane bounded
- Recognized by T-cells with antigen as a result of cell to cell interaction
- For an immune response, peptide must be bounded with MHC (One level control)

Important Aspects of MHC

Class II MHC

- Mature T-cell must have a TCR which recognize MHC bounded peptide (second level control)
- Cytokines increase the expression of MHC
- Peptides from cytosol: Class I MHC
- Peptides from vesicle: Class II MHC
- Polymorphism in MHC

Immunology

Important Aspects of MHC

(Merged by
Talha Waraich
For help What's
App or Telegram at
03488485459)

Role of MHC in Tissue Matching

Tissue Matching

- Prospective donor & recipient tested for compatibility prior to transplantation
- MHC also called as Human leucocyte antigen (HLA)
- HLA molecules are the primary target of immune responses to allogeneic transplants

Role of MHC in Tissue Matching

Tissue Matching

- HLA molecules are highly diverse in human population
- HLA typing or tissue matching detects & classifies this diversity
- Extensive polymorphism of HLA: less chances of HLA matched donors for transplants
- Appx.25% of siblings inherit same HLA type

END

Immunology

Self MHC Restriction

Self MHC Restriction

- For an appropriate immune response, T-cells should recognize & respond to foreign Ag
- Foreign Ag presented to T-cells must be self MHC
- Ag: Foreign
- MHC: Self

Self MHC Restriction

- Cytotoxic T-Cells recognize Ag in context of Self Class I MHC
- Helper T-Cells recognize Ag in context of Self Class II MHC
- This process whereby T-cells become restricted to recognize self-MHC occur in thymus

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Differences B/W Monoclonal & Polyclonal Antibodies

Monoclonal Ab

- Antibody from a single antibody producing B-cell
- Also able to bind with single & unique epitope
- Mainly consist of single subtype of IgG e.g IgG1, IgG2 & IgG3

Differences B/W Mono & Polyclonal Ab

Polyclonal Ab

- Collection of Antibodies from a different antibody producing B-cell
- Also able to bind with multiple epitopes on a same antigen
- Obtained in serum with antibodies having different affinities
- Mainly belong to IgG class

Immunology

Generation of Polyclonal Antibodies

Generation of Polyclonal Antibodies

Polyclonal Ab

- The general procedure for generation of polyclonal antibodies include following phases
- Antigen preparation/Antigen & Adjuvant conjugate
- Antigen Immunization
- Antibody Purification from animal's serum
- Antibody Validation

Generation of Polyclonal Antibodies



<https://www.novoprolabs.com/custom-polyclonal-antibodies/>

Generation of Polyclonal Antibodies

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<https://www.novoprolabs.com/custom-polyclonal-antibodies/>

Role of Adjuvants in Polyclonal Antibodies Production

Role of Adjuvants/Polyclonal Antibodies

Adjuvants

- Potentiates immune response to antigen
- Modulates towards desired immune response
- Also have undesirable effects like toxicity
- Selection would made in order to get maximum immunostimulation

Role of Adjuvants/Polyclonal Antibodies

Adjuvants

- For polyclonal antibodies production following adjuvants are used
- Freund's Adjuvants
- Freund's complete adjuvant (FCA): water-in-oil emulsion containing antigen with heat killed *Mycobacterium tuberculosis*
- Stimulate both humoral & Cell mediated immune response for Ab

Role of Adjuvants/Polyclonal Antibodies

Adjuvants

- Freund's Incomplete adjuvant (FIA): water-in-oil emulsion containing antigen without heat killed *Mycobacterium tuberculosis*
- Used as booster antigen dose
- Freund's adjuvant is used by mixing with equal parts of antigen

Role of Adjuvants in Polyclonal Antibodies Production

Role of Adjuvants/Polyclonal Antibodies

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- Freund's adjuvant is used by mixing with equal parts of antigen

Production of Monoclonal Antibodies

Production of Monoclonal Antibodies

Monoclonal Antibodies

- Production of monoclonal antibodies is done using Immortal clone of cell with single Ab specificity
- Immortal cell: Not proliferate indefinitely
- Fusion of normal Ab producing cell with an appropriate B-cell tumor line (Hybrid Cell)
- Large production of Ab

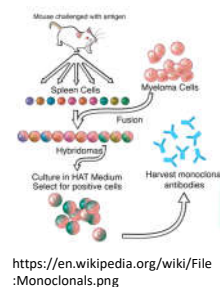
Hybridoma Formation

Hybridoma Formation

Hybridoma

- Hybrid cell line: Fusion of Lymphoid tumor cell with normal B-Lymphocyte with single specificity
- Dual function: Immortality of tumor cell & production of Ab with single specificity
- Principle involves
- Injection of mice with an antigen

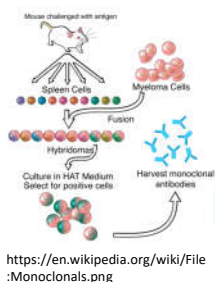
Hybridoma Formation



Hybridoma

- Isolation of B-lymphocyte from spleen of mice
- Fusion of that B-lymphocyte with lymphocyte immortal cell i.e Myeloma cell
- Fused cell: Hybridoma
- Selection Hybridoma cell: Using HAT (Hypoxanthine, aminopterin, thymidine)

Hybridoma Formation



Hybridoma

- Only hybridoma cell can grow on HAT medium
- Propagation of hybridoma cell: for large amount of monoclonal Ab
- Dual function: Ab production plus exaggerated longevity

Usage of Monoclonal Antibodies
(Merged by
Talha Waraich
For help What's App or Telegram
at
03488485459)

Usage of Monoclonal Antibodies

- Usage in prevention, diagnosis & treatment of disease
- Immunophenotyping of Immune cells: Monoclonal Ab used against cell surface molecules like Cluster of Differentiation (CD)
- Typing of various tumors e.g Leukemia
- Immunohistochemistry of solid tumors

Usage of Monoclonal Antibodies

- Use in Immunotherapy: Against various tumor marker e.g CD20 B-Cell lymphoma
- Use in Fluorescence Activated cell sorting (FACS)

Immunology

T-Lymphocyte Receptors, Maturation, Activation & Differentiation

Immunology

Structure of T-Cell Receptor (TCR)

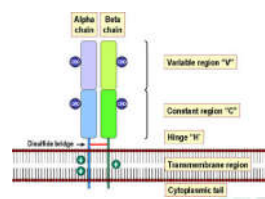
Structure of TCR

TCR

- Surface receptor for binding with Ag bounded with MHC
 - Like immunoglobulin: Part of Ig Superfamily
 - TCR is a heterodimer surface receptor
 - Composed of two chains of equal length
1. A Chain
 2. B Chain

Structure of TCR

TCR

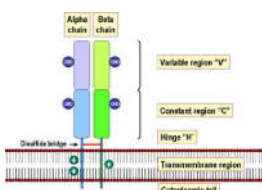


<http://www.microbiologybook.org/bowers/tcr.jpg>

- Short Cytoplasmic tail: Not sufficient for signal transduction
- Transmembrane region: for anchoring cell membrane (due to hydrophobic amino acids)
- Disulfide bridge B/W two chains
- Both chains contain carbohydrate moiety

Structure of TCR

TCR



<http://www.microbiologybook.org/bowers/tcr.jpg>

- Both chains contain constant (C) & Variable (V) Regions
- Variable regions: Hypervariable regions in both chains
- Specificity of TCR: Variable region of both chains
- Each TCR carry single specificity

Immunology

Diversity of T-Cell Receptor (TCR)

Diversity of TCR

- Genetic basis: Diverse array of TCR
- Joining of Various Gene segments
- T-cell maturation: Thymus
- Germ line gene for α chain: V & J segments
- Germ line gene for $\alpha\beta$ chain: V, D & J segments
- Combinational diversity: For V, D & J

Diversity of TCR

- Specificity of TCR: Combination of α & β chains
- Small population T-cells: Carrying TCR with γ (Gamma) & δ (Delta) chains
- T-cells predominates mucosal epithelium: for certain bacteria & viral antigens

Diversity of TCR

- Repertoire of $\gamma\delta$ T-cells : Joining gene segment
- Gamma chain gene contain V & J segment
- Delta chain germline gene: V, D & J segments
- Combinational diversity
- $\gamma\delta$ T-cells recognize antigens independent of MHC association